

DiffCSE: Difference-based Contrastive Learning for Sentence Embeddings

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Background

Sentence embedding = converting a sentence into a dense vector.

Example:

- "The cat is sleeping."

→ [0.21, -0.53, 0.92, ...]

- Desired property:

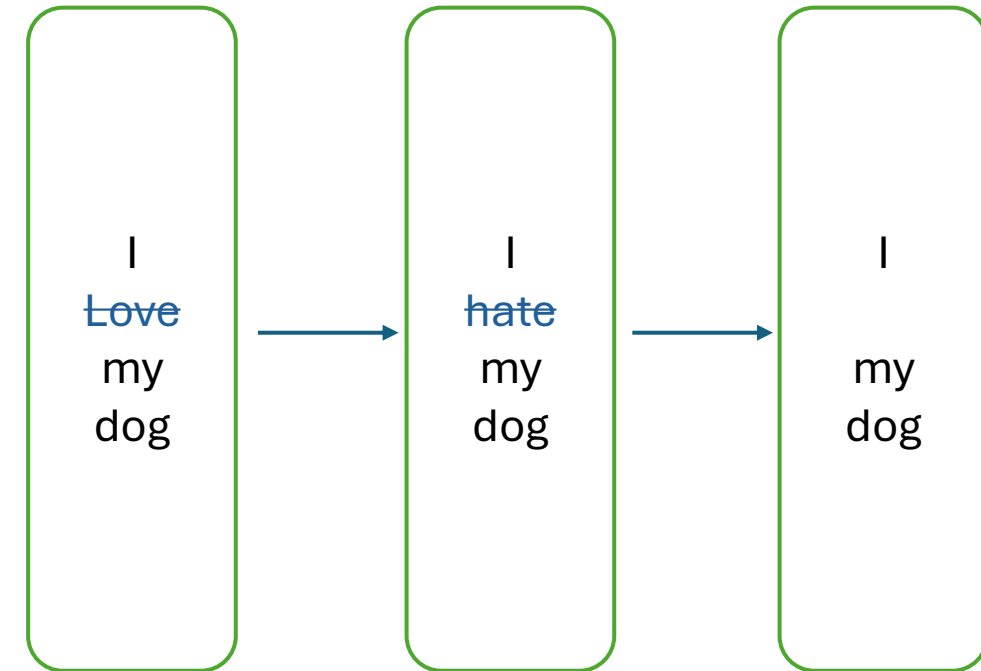
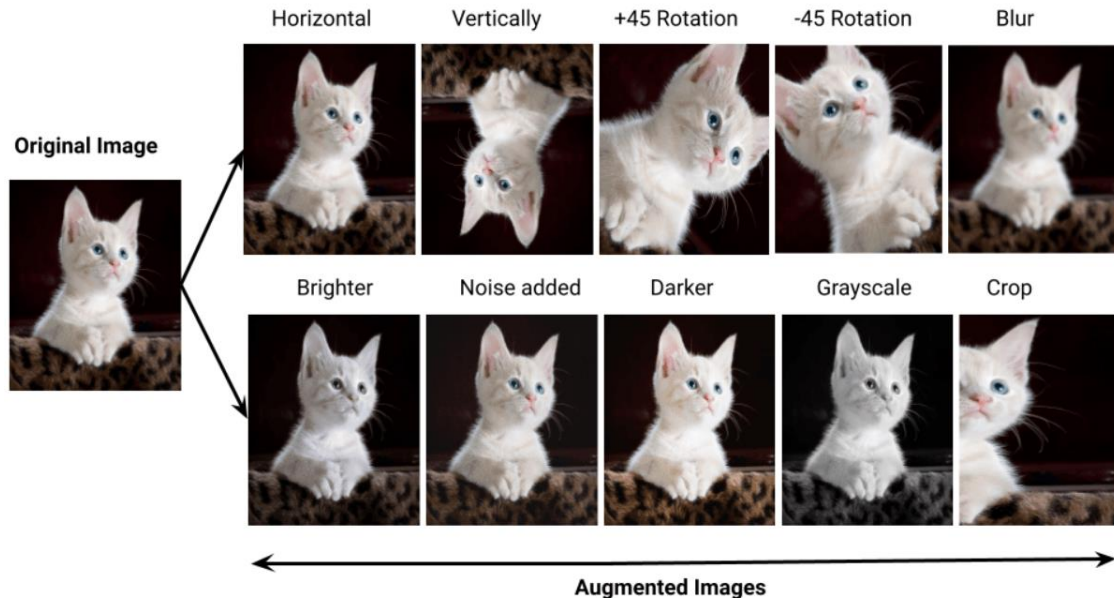
"The cat is sleeping." | "The kitten is asleep." → vectors should be close

Applications:

- Semantic Textual Similarity (STS)
- Search
- Retrieval
- Question Answering
- Clustering

Contrastive learning: From CV-NLP

- A training technique that learns representations by pulling similar samples closer together and pushing dissimilar samples farther apart in the embedding space.



Problem with SimCSE

- SimCSE (Simple Contrastive Learning of Sentence Embeddings) generates two views of the same sentence using different dropout masks. However, dropout only changes internal neurons and does not introduce meaningful semantic variation. DiffCSE addresses this by creating corrupted sentences and learning the semantic differences between the original and corrupted versions.
- Invariant contrastive learning: Different views of the same sentence should have the same meaning representation.

SimCSE

I love machine learning.



Dropout #1 $\rightarrow z_1$

I love machine learning.



Dropout #2 $\rightarrow z_2$

Difference: random neurons

Introducing DiffCSE

- DiffCSE, an unsupervised contrastive learning framework for learning sentence embeddings.
- DiffCSE learns:
 - Similarity via contrastive learning
 - Semantic differences via difference prediction
- The DiffCSE model is trained in a fully unsupervised manner using a large corpus of raw text from Wikipedia
- It achieved state-of-the-art performance among unsupervised sentence embedding methods. On Semantic Textual Similarity (STS) tasks, it outperformed SimCSE by 2.3 points.

Equivariant contrastive learning:

- invariant to harmless transformations (like dropout)
- sensitive to meaningful transformations (like word replacement)

DiffCSE

I love machine learning.

Corrupted:

I [MASK] machine learning.

Difference: actual missing word

* Now the model must understand meaning.

DiffCSE Architecture

1. Generator (MLM)

- Creates corrupted sentences.
- Example: Original: The cat sits on the mat.
Corrupted: The [MASK] sits on the mat.

2. Encoder

- BERT : Converts sentences into embeddings.
- Sentence -> Encoder -> Embedding

3. Discriminator

- Receives: Receives h (original sentence embeddings and the edited sentence x') and learns What was changed!
- This creates the "difference-based" supervision.

Training Objective

Two losses are used.

A. Contrastive Loss

- Similar to SimCSE.
- Pull positive pairs together and push negatives apart.
- Use InfoNCE loss.

B. Difference Prediction Loss

- The discriminator predicts:
- Which tokens were modified?
- This forces semantic reasoning.

Datasets Used for Training

SNLI (Stanford Natural Language Inference) dataset is used as the training corpus.

Each SNLI sample contains:

- Premise
- Hypothesis
- Label (neutral, contradiction and entailment)

Number of rows: 5,50,152

Split (3) train · 550k rows		
Search this dataset		
premise string · lengths	hypothesis string · lengths	label class label
		
A person on a horse jumps over a broken down airplane.	A person is training his horse for a competition.	1 neutral
A person on a horse jumps over a broken down airplane.	A person is at a diner, ordering an omelette.	2 contradiction
A person on a horse jumps over a broken down airplane.	A person is outdoors, on a horse.	0 entailment
Children smiling and waving at camera	They are smiling at their parents	1 neutral
Children smiling and waving at camera	There are children present	0 entailment
Children smiling and waving at camera	The kids are frowning	2 contradiction
< Previous 1 2 3 ... 5,502 Next >		

Datasets Used for Evaluation

The paper evaluates mainly on: STS Benchmark (Semantic Textual Similarity)

Common datasets include:

- STS12, STS13, STS14, STS-B etc.
- We used: **STS-B** (STS Benchmark) :
- Contains roughly:

Train 5,749 pairs

Validation 1,500 pairs

Test 1,379 pairs

Split (3) train · 5.75k rows		
Search this dataset		
sentence1 string · lengths 	sentence2 string · lengths 	score float64 
A plane is taking off.	An air plane is taking off.	1
A man is playing a large flute.	A man is playing a flute.	0.76
A man is spreading shredded cheese on a pizza.	A man is spreading shredded cheese on an uncooked pizza.	0.76
Three men are playing chess.	Two men are playing chess.	0.52
A man is playing the cello.	A man seated is playing the cello.	0.85
Some men are fighting.	Two men are fighting.	0.85
A man is smoking.	A man is skating.	0.1
The man is playing the piano.	The man is playing the guitar.	0.32
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Results

Evaluated on 7 STS benchmarks.

- DiffCSE consistently outperformed SimCSE.
- STS Score:

SimCSE-BERT : 76.25

DiffCSE-BERT : 78.49

Qualitative retrieval examples:

Query: This is not a problem.

SimCSE: This is a big problem. [*top mapped result*]

DiffCSE: I don't see why this could be a problem. [*top mapped result*]

Strengths & Limitations

Strengths

- Better semantic supervision : Learns actual meaning differences.
- Self-supervised : No manual labels needed.
- Strong STS performance: Beats SimCSE.
- Works with BERT/RoBERTa : Easy to adapt.

Limitations

- More complex than SimCSE: Need:
 - Generator, Discriminator, Extra losses
- More computation : Training cost increases.
- Failed on double negation sentences [i think that is not a bad idea] DiffCSE generated; it is not a good idea.
- Generator vs Discriminator